

1 ■ Quadric Interpolant

Exercise 1.1. Find the second degree parabola passing through the given points:

- (a) $(0, 2), (1, 5), (2, 14)$
- (b) $(-1, 2), (3, 14), (0, -1)$
- (c) $(1, -5), (-1, 1), (2, 7)$

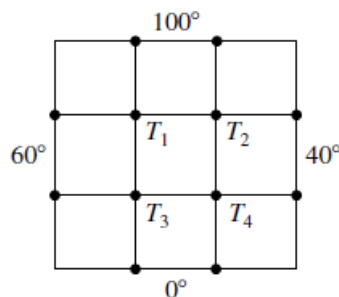
Hint. Let $y = ax^2 + bx + c$ be the parabola passing through the given points. Substitute the coordinates of the points in it, derive an appropriate system of equations, and then solve it.

Answers.

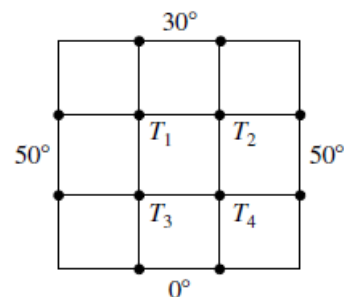
- (a) $y = 3x^2 + 2$
- (b) $y = 2x^2 - x - 1$
- (c) $y = 5x^2 - 3x - 7$

2 ■ Temperature Distribution

Exercise 2.1. A square plate is perfectly insulated on its top and bottom so that the only heat flow is through the plate itself. The four edges are held at various temperatures. The temperature at an interior point on the plate is the average of the temperatures at its four compass-point neighbors, to the west, north, east, and south. Find out the temperatures at the interior points $T_i, i = 1, 2, 3, 4$ for the plates shown in the following figures:



(a) Plate 1



(b) Plate 2

Figure 1

Answers.

- (a) $T_1 = 65^\circ, T_2 = 60^\circ, T_3 = 40^\circ, T_4 = 35^\circ$

3 ■ Global Positioning System

Exercise 3.1. Determine the planar location (x, y) of a GPS receiver, using coordinates of the centers and radii for the three circles given in the following tables:

Circle	Center	Radius
1	$(-15, 20)$	25
2	$(5, -12)$	13
3	$(9, 40)$	41

(a) Table 1

Circle	Center	Radius
1	$(-10, 13)$	25
2	$(10, -19)$	13
3	$(14, 33)$	41

(b) Table 2

Figure 2

4 ■ Traffic Flow Model

Exercise 4.1. Use Gauss-Jordan elimination method to find the coefficients solve the traffic flow model, represented by the the following diagrams:

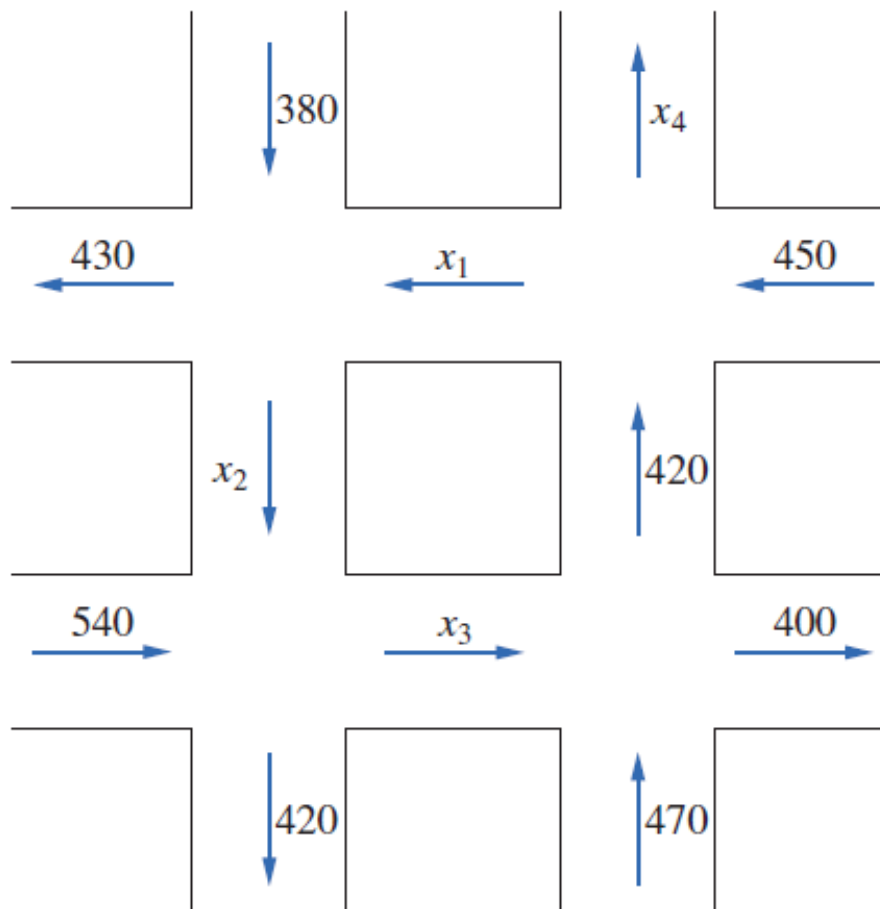


Figure 3: Model 1

Exercise 4.2. Find the coefficients in the traffic flow model:

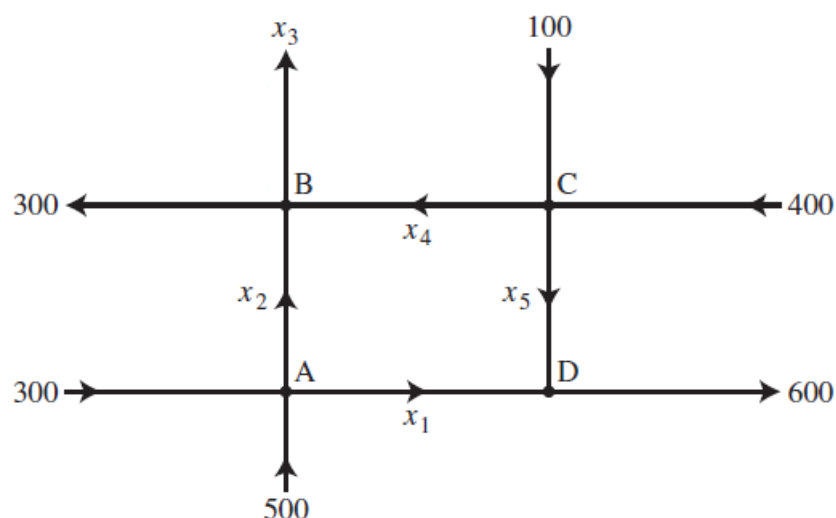


Figure 4: Model 2

5 Current Flow in Networks

Exercise 5.1. Find the amount of each current for the following networks:

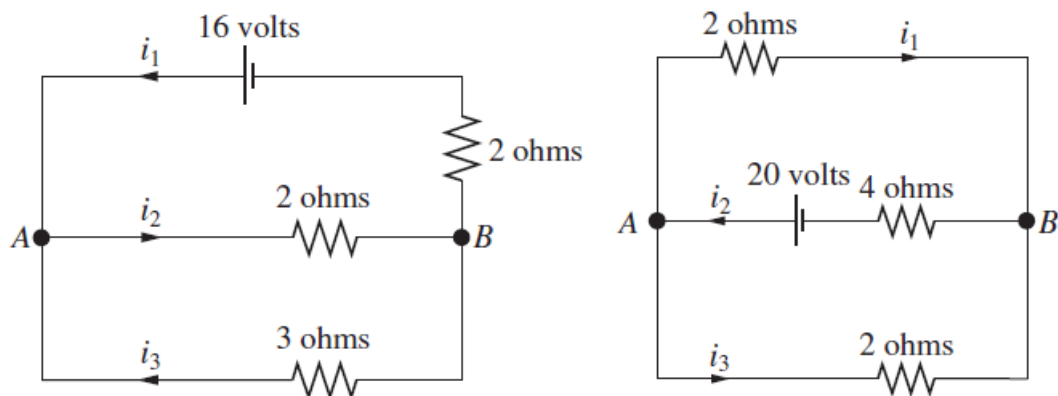


Figure 5: Network Currents

Exercise 5.2. Find the currents in the following electrical networks:

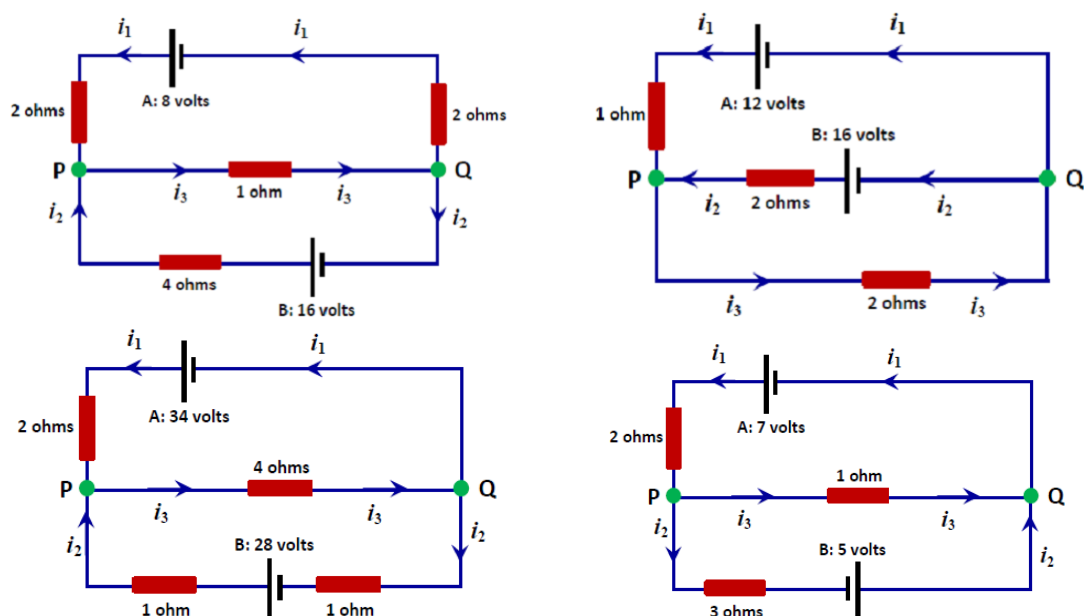


Figure 6: Network Currents

6 Chemical Balance Equations

Exercise 6.1. Use Gauss-Jordan elimination method to balance the chemical equations in the following reactions:

- Sodium hydroxide (NaOH) reacts with sulfuric acid (H_2SO_4) to form sodium sulfate (Na_2SO_4) and water (H_2O). The chemical equation is $\text{NaOH} + \text{H}_2\text{SO}_4 \longrightarrow \text{Na}_2\text{SO}_4 + \text{H}_2\text{O}$
- The reddish brown rust layer (Fe_2O_3) is formed when iron (Fe) reacts with oxygen (O_2). The chemical equation is $\text{Fe} + \text{O}_2 \longrightarrow \text{Fe}_2\text{O}_3$.
- Ethane gas (C_2H_6) burns in oxygen (O_2) to give carbon dioxide (CO_2) gas, and the steam condensed as water droplets (H_2O). The chemical equation is $\text{C}_2\text{H}_6 + \text{O}_2 \longrightarrow \text{CO}_2 + \text{H}_2\text{O}$.

- (d) Potassium chlorate (KClO_3) decomposes into potassium chloride (KCl) and oxygen (O_2) when heated. The chemical equation is $\text{KClO}_3 \longrightarrow \text{KCl} + \text{O}_2$.
- (e) When copper (Cu) is combined with nitric acid (HNO_3), a redox reaction occurs that gives copper nitrate ($\text{Cu}(\text{NO}_3)_2$), nitrogen dioxide (NO_2) and water (H_2O). The chemical equation is $\text{Cu} + \text{HNO}_3 \longrightarrow \text{Cu}(\text{NO}_3)_2 + 2\text{NO}_2 + 2\text{H}_2\text{O}$
- (f) Propane Combustion: $\text{C}_3\text{H}_8 + \text{O}_2 \longrightarrow \text{CO}_2 + \text{H}_2\text{O}$
- (g) Fermentation of Sugar: $\text{C}_6\text{H}_{12}\text{O}_6 \longrightarrow \text{CO}_2 + \text{C}_2\text{H}_5\text{OH}$
- (h) Photosynthesis: $\text{CO}_2 + \text{H}_2\text{O} \longrightarrow \text{C}_6\text{H}_{12}\text{O}_6 + \text{O}_2$